

KOZIENICE, Poland

WMO: 124880

Lat: 51.5647N

Lon: 21.5436E

Elev: 126

StdP: 99.82

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-13.2	-19.4	0.7	-16.5	-16.0	0.9	-12.2	9.1	2.7	8.3	1.1	1.6	110	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	30.9	20.7	28.8	19.8	27.1	19.0	21.7	28.2	20.8	27.0	19.9	25.4	3.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.5	24.1	18.7	13.7	23.1	17.9	13.1	22.2	63.8	28.5	60.5	27.1	57.3	25.4	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.7	6.8	6.0	DB	-20.2	33.7	5.2	1.5	-24.0	34.8	-27.0	35.7	-29.9	36.6	-33.7	37.7	(4)
			WB	-20.4	23.4	4.9	1.0	-23.9	24.2	-26.8	24.8	-29.5	25.4	-33.1	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-2.6	-0.9	3.2	8.9	13.6	17.3	19.4	18.7	13.7	8.7	4.3	0.1	(6)
(7)		DBStd	8.66	5.93	5.36	4.81	4.03	3.64	3.36	3.16	3.09	3.58	4.27	4.08	5.08	(7)
(8)		HDD10.0	1552	390	304	216	67	9	0	0	0	8	76	176	307	(8)
(9)		HDD18.3	3637	648	538	470	283	151	59	26	33	144	298	421	565	(9)
(10)		CDD10.0	1098	0	0	4	34	121	218	292	271	119	36	4	0	(10)
(11)		CDD18.3	142	0	0	0	1	5	27	59	44	6	0	0	0	(11)
(12)		CDH23.3	1661	0	0	0	21	105	334	625	498	78	1	0	0	(12)
(13)		CDH26.7	479	0	0	0	3	16	83	204	158	14	0	0	0	(13)
(14)	Wind	WSAvg	2.8	3.3	3.2	3.3	2.9	2.6	2.5	2.4	2.2	2.3	2.7	2.9	3.1	(14)
(15)	Precipitation	PrecAvg	598	33	32	37	42	64	66	93	62	53	40	39	36	(15)
(16)		PrecMax	748	85	68	85	108	143	152	290	202	127	101	81	83	(16)
(17)		PrecMin	445	4	8	8	1	20	30	10	10	4	5	2	7	(17)
(18)		PrecStd	80	18	14	17	26	32	32	61	37	35	28	15	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	12.1	18.1	25.8	28.5	31.8	33.5	28.7	22.3	15.7	11.3	(19)	
(20)		MCWB	7.8	8.5	10.7	15.6	18.9	21.5	20.9	21.0	18.8	16.2	12.2	9.0	(20)	
(21)		2%	DB	7.9	9.5	14.9	22.3	26.2	28.9	31.1	30.5	25.5	19.9	13.4	9.4	(21)
(22)		MCWB	6.0	7.0	9.1	13.9	17.7	20.2	20.7	20.7	17.7	14.9	10.7	7.7	(22)	
(23)		5%	DB	6.2	7.7	12.6	19.3	24.0	27.0	28.8	28.2	23.3	17.6	11.8	7.7	(23)
(24)		MCWB	4.7	5.6	8.0	12.1	16.6	18.9	19.8	19.4	16.7	13.7	9.7	6.3	(24)	
(25)		10%	DB	4.6	6.0	10.2	17.1	21.9	25.0	26.8	26.0	21.0	15.5	10.3	6.1	(25)
(26)		MCWB	3.3	4.3	6.9	11.0	15.3	17.7	19.1	18.6	15.3	12.7	8.8	5.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	8.8	11.6	16.0	20.0	22.6	22.9	19.7	17.0	12.7	9.8	(27)	
(28)		MCDWB	9.6	11.1	17.4	25.1	26.6	28.8	30.1	30.2	26.7	21.0	14.9	11.1	(28)	
(29)		2%	WB	6.2	7.3	9.9	14.5	18.5	21.2	21.8	21.4	18.3	15.4	11.3	7.8	(29)
(30)		MCDWB	7.6	9.3	14.1	21.0	24.7	27.6	28.0	28.2	24.0	19.0	13.0	9.0	(30)	
(31)		5%	WB	4.9	5.7	8.5	13.0	17.2	19.9	20.8	20.4	17.2	14.1	10.0	6.3	(31)
(32)		MCDWB	6.1	7.4	11.6	18.2	22.9	25.3	26.8	26.7	22.1	17.1	11.5	7.5	(32)	
(33)		10%	WB	3.3	4.3	7.2	11.6	16.0	18.7	19.9	19.4	16.0	12.8	8.8	5.1	(33)
(34)		MCDWB	4.4	5.8	9.8	15.9	20.6	23.4	25.1	24.5	19.9	15.3	10.2	6.1	(34)	
(35)	Mean Daily Temperature Range		MDBR	4.9	6.0	8.3	11.2	11.7	11.5	11.5	11.9	10.7	8.8	5.6	4.7	(35)
(36)		5% DB	MCDBR	5.3	8.1	12.6	15.9	15.8	15.1	15.5	15.7	14.9	12.3	8.5	5.7	(36)
(37)			MCWBR	4.4	5.8	7.5	8.2	8.2	7.3	6.2	6.9	7.6	7.7	6.1	4.4	(37)
(38)			MCDBR	5.2	7.4	11.1	14.2	14.1	13.3	13.1	13.8	13.2	11.2	7.4	5.3	(38)
(39)		5% WB	MCWBR	4.5	5.6	7.1	7.8	7.7	7.0	5.9	6.3	7.0	7.2	5.6	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.321	0.348	0.384	0.437	0.422	0.419	0.447	0.440	0.403	0.372	0.357	0.317	(40)	
(41)		taud	2.237	2.221	2.200	2.113	2.220	2.260	2.196	2.214	2.280	2.333	2.286	2.278	(41)	
(42)		Ebn at Noon	666	755	799	797	832	838	804	787	773	722	617	608	(42)	
(43)		Edh at Noon	70	93	115	141	133	129	136	127	106	83	68	58	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.73	1.46	2.68	4.08	5.13	5.50	5.12	4.55	3.23	1.84	0.80	0.55	(44)	
(45)		RadStd	0.14	0.24	0.35	0.55	0.55	0.48	0.61	0.35	0.49	0.28	0.08	0.09	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (21 neighbors)	+0.54	N/A	N/A	+0.75	+0.43	N/A	N/A	-168	+82	+31	(47)

Nomenclature: See separate page